Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026, 07:59:59 MYT (UTC+8)

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	CodeFirst
Project / Solution Name	Cafe Replenishment
Name (Leader)	Fwn Jiarui
Email & Phone (Leader)	1486866989@qq.com/+60 175441714
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://www.cafe-replenishment-demo.com/demo/
Code Repository URL (public / access granted to judges)	**https://github.com/wanblairr/code-first — public repository, Apache License 2.0 LICENSE file at the repository root, main branch only. Committee account(s) added as collaborators; access is managed by the team leader at Settings → Collaborators.**

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end–to–end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The public browser demo combines a selected preparation date, Kuala Lumpur weather from Open-Meteo (or a clearly labelled simulated scenario), and a verified 2026 Kuala Lumpur public-holiday calendar with deterministic synthetic sales history. A project-developed ridge-regression forecast estimates portions for espresso, latte and oat latte; the demo uses an explicit simulated product mix for its wider 20-item menu. CSV recipe/BOM data converts menu demand into ingredient use. The replenishment calculation considers remaining stock, confirmed inbound quantities, safety stock and supplier pack sizes. The manager can review and adjust quantities, then explicitly approve a simulated purchase draft. Core demo state and activity history stay in the visitor's browser. Optional integrations are separate: Gmail can save an approved draft; a visitor-selected Runpod endpoint can provide a paid AI review after explicit approval; WhatsApp test messages require a separate send confirmation. Hosting uses Netlify static delivery with serverless functions and bounded Netlify Blobs records for optional integrations. No purchase, payment or email send is performed automatically. The published backtest uses synthetic data only and has not been validated for a real café. The module is designed to plug into Orbyte as a reusable hub component serving many SMEs.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Native HTML, CSS and JavaScript (public browser client)	Interface, demo workflow and local state; only native assets were identified in the served page	N/A — native web standards, not a licensed third-party library

Name & Version	Purpose in Project	License
openpyxl 3.1.5; et_xmlfile (bundled with openpyxl)	Excel import/export in the local workbench (BOM, sales history, purchase drafts). No npm packages are used server-side; the Netlify Functions use only the Node.js runtime built-ins.	MIT (both)

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Open-Meteo Forecast API	Live weather for fixed Kuala Lumpur coordinates and selected date; simulation modes are labelled	Public/free access for non-commercial use with attribution (Open-Meteo terms)
Google Identity Services + Gmail API (gmail.compose)	Optional visitor-owned Google connection; saves an approved supplier draft only; app has no email-send operation	User-authorized Google account; not sponsor-provided
Runpod API / visitor-supplied endpoint	Optional paid AI review; only approved synthetic forecast/context payload is sent	Visitor-owned endpoint/key; paid usage; not sponsor-provided
Meta WhatsApp Cloud API	Optional advanced test; user-provided Meta setup; sending requires separate confirmation	User-owned account/permissions; tier or charges depend on account; not sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Netlify static hosting	Hosts the public demo website and browser assets	Free (Starter) plan; global CDN edge, region auto-assigned

Service & Provider	Purpose in Project	Plan / Region
Netlify Functions + Netlify Blobs	Optional Runpod/WhatsApp server-side requests and hashed duplicate-send/job guards; no raw keys or raw message contents are described as persisted	Free (Starter) plan; global CDN edge, region auto-assigned
Browser localStorage (not a cloud service)	Stores the public demo's simulated café state and activity history in the visitor's browser profile	Local to browser; cleared by demo reset where applicable
Separate local owner workbench	Separate operator-controlled local runtime; not the public browser demo deployment	Operator-controlled local machine; not a cloud service

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Cafe Replenishment ridge regression, espresso (context:0.1)	Project-developed model; public model metadata at /demo/model.json	Self-trained/fitted on synthetic history	Project model; no separate model license stated
Cafe Replenishment ridge regression, latte (context:1.0)	Project-developed model; public model metadata at /demo/model.json	Self-trained/fitted on synthetic history	Project model; no separate model license stated
Cafe Replenishment ridge regression, oat latte (context:1.0)	Project-developed model; public model metadata at /demo/model.json	Self-trained/fitted on synthetic history	Project model; no separate model license stated
None bundled — the optional AI review calls a model the visitor supplies through their own RunPod endpoint; no model is embedded or pre-configured by the team	Visitor-supplied RunPod endpoint	Pre-trained (visitor-selected at runtime)	Governed by the visitor's chosen provider terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic sales and simulated historical weather for model fit/backtest (180 days; 540 sales rows)	Public metadata: https://www.cafe-replenishment-demo.com/demo/model.json	Synthetic; generated for demo (training/validation/56-day chronological holdout)	Project-generated; no separate dataset license stated	N
Visitor-date synthetic history generated relative to selected target date	Same public model metadata URL	Synthetic; regenerated deterministically; no real business records	Project-generated; no separate dataset license stated	N
Demo product catalog, CSV recipes/BOM, stock, safety levels and suppliers	Public browser demo and model metadata	Synthetic / simulated demonstration data	Project-generated; no separate dataset license stated	N
Kuala Lumpur 2026 public-holiday calendar	Source list in model.json; Malaysian government calendar, gazette and agency references	Public official sources, curated for the demo calendar	Public official Malaysian government sources; cited in model.json with attribution	N
Selected-date live weather response	Open-Meteo API; fixed Kuala Lumpur coordinates	Public external data feed, used at runtime	Open-Meteo public API terms; attribution provided in model.json	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex and ChatGPT	Engineering revisions and code assistance; debugging and review; technical documentation and presentation material. The release NOTICE records Codex / ChatGPT assistance. Model versions are managed by OpenAI and were not pinned by the team.	Major (extensive use)
Replit Agent (built into Replit Core; model version is not exposed in the Replit UI)	Code review and bug fixes on the replenishment logic, and the client for OpenAI-compatible endpoints, inside the Replit Project.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members review AI-assisted code and documentation against project requirements, inspect changes, and retain approval of design and release decisions. Verification records report 619/620 Python tests passed (one OS-specific test skipped), 40/40 local UI tests passed, and 332/332 public-source tests passed. These checks cover engineering and synthetic demo behavior, not real-café accuracy. Before submission, the team leader verifies claims, licenses, and data handling. The team remains accountable for all submitted code and content; AI suggestions are not accepted automatically.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core credit (US$20 per participant)	**Replit Core was the team's development environment. The browser client, the Netlify Functions and the model-fitting scripts were written, reviewed and debugged inside a Replit Project, and Replit Agent was used for code review and bug fixes on that code base. Replit is not part of the deployed build; the live demo is served by Netlify. Proof of use: SDC2026_CodeFirst_Replit_POU.pdf — the Replit Agent session on this code base, plus the credit-exhausted screen after the US$20 SDC allocation was consumed.**

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	US$25 team credit (GPU compute)	**RunPod was used for the optional GPU path. A GPU pod (extreme_ivory_sheep; 1x GPU, 24 vCPU, 251 GB RAM; image runpod/pytorch:1.4.1-cu1281-torch280-cu1284; region US-CA-2) was used to load and test open-source models that a visitor may select for the advisory review. Billed at US$7.99/hr. The core replenishment workflow needs no GPU. Proof of use: SDC2026_CodeFirst_Runpod_POU.pdf — pod details, runtime and the Connect tab.**

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was used as the development environment and for AI-assisted code review, and is not essential to the deployed build, which runs on Netlify. RunPod was used only for the optional GPU path (loading and testing an open-source model); the core replenishment workflow requires no GPU and could run on any GPU provider. No other sponsor resource was used.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Public demo needs no account. Simulated stock, approvals, drafts and activity history stay in browser localStorage; the host may process request metadata. Weather requests send the selected date and fixed Kuala Lumpur coordinates to Open-Meteo; visitor location is not used. Optional Gmail sends account identity and recipient, subject and body to Google to create a draft; inbox bodies are not read. OAuth tokens stay in tab memory. Optional RunPod review sends previewed synthetic forecast and weather/holiday context, plus the visitor-supplied endpoint and key, through Netlify Functions; the app does not persist them. Optional WhatsApp sends the user's token, numbers, template and approved summary to Meta/360dialog after separate confirmation; Blobs stores only hashes and status. The local workbench keeps recipes, stock, suppliers, forecasts and drafts on the operator's computer; Windows tokens use DPAPI. Browser records persist until reset; local records until deletion.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The forecast is trained and evaluated on deterministic synthetic data only; the published 56-day holdout does not establish real-café accuracy. It forecasts portions and applies a simulated mix for the wider menu. Weather availability and the selected holiday calendar may be stale or incomplete, while supplier prices, pack sizes, stock and safety levels need owner confirmation. The optional Runpod model is visitor-selected and may be unavailable, costly, or produce unsuitable advice; usage caps and timeouts do not guarantee completion. Open-Meteo, Google, Meta, 360dialog and Netlify are external dependencies with their own availability and terms. Mitigations include clear simulated-data labels, visible payload previews, explicit owner review and approval, separate confirmation for message sending, no automatic purchasing/email sending, and blocked ambiguous/retry states. Human review remains necessary.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: _____ **Name: Fwn Jiarui** **Date: 24 October 2026**

